G.A.T.E.S. — Gate 1

What it checks, what it measured, and what changed when it was switched on

Generated 2026-08-15 · gates @ main · 249 gate tests, 63 host integration tests · every number below is a recorded measurement or a cited published figure

1. The defect, in one line of the host scaffold

```
def execute_code(code_str, timeout=60, MAX_LEN=1000):
    ...
    except Exception as e:
        output_capture.write(f"[CODE EXECUTION ERROR]: {str(e)}
")   # appended AFTER the prints
    return output_capture.getvalue()[:MAX_LEN]                        # ...then sliced off
```

The writing agent's entire experimental record was 1,000 characters, and the crash marker was appended after the program's own output — so on any run that printed more than that, the marker fell off the same slice and the solver's only crash test never fired.

measured in the archived run	value
Reward score on a run raising `NameError` every attempt	**0.95 → 0.98 → 1.0**
Test accuracy reported in the paper	81.60% (never measured)
Speedup reported	13.61× (never measured)
Ablation collapse reported	39.20% (never measured)
Archived runs usable of seven	**1**

2. The run: a data-efficiency study, with and without Gate 1

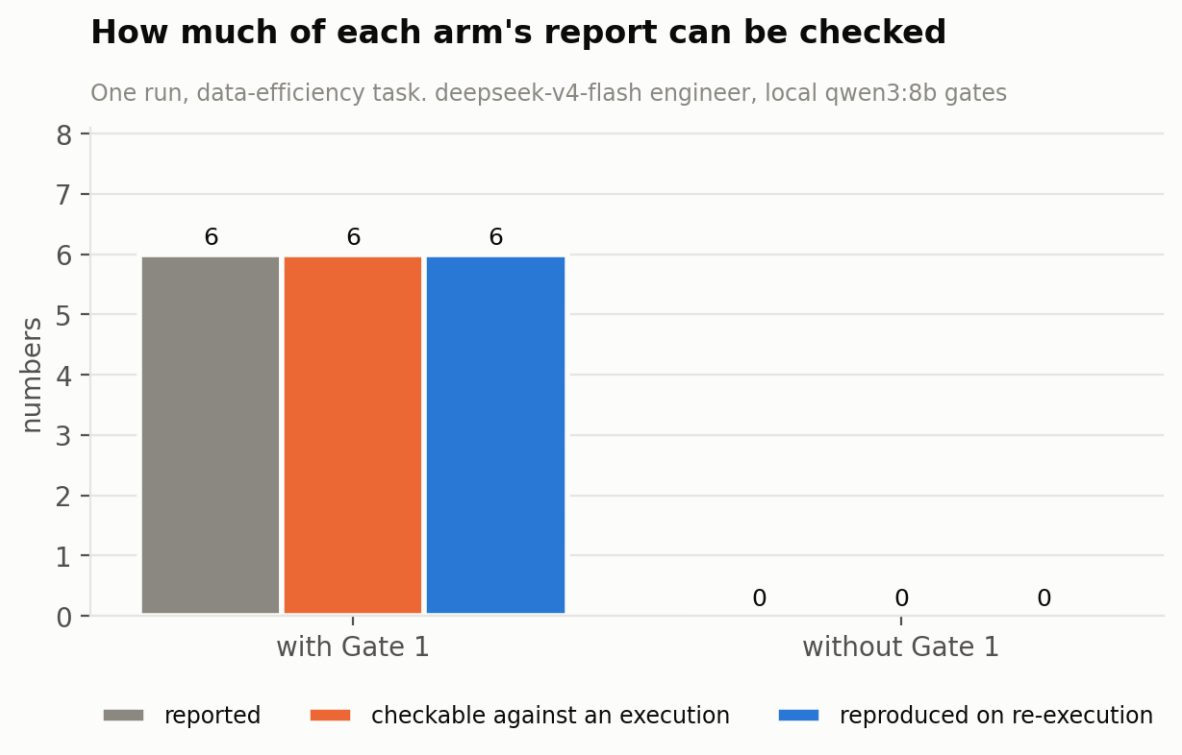
Research question given to the agent: how much labelled data does a classifier need? Make a synthetic 3-class dataset, train multinomial logistic regression by gradient descent on 25% of the training samples and on all of them, and report the two test accuracies and their ratio — three values.

A tractable proxy for the question asked of large language models: training one locally is not possible here, so the same scaling-with-data question is put to a classifier the agent can actually implement, and this report says so rather than implying otherwise.

Models. The gates run on a local `qwen3:8b`, as specified. The engineer runs on `qwen2.5-coder:7b`, and the reason is itself a measurement: a reasoning model at this scale spends its whole token budget in the reasoning channel and returns `finish_reason="length"` with *empty content* — reproduced at 2,500, 3,000, 6,000 and 8,000 token ceilings, with `/no_think`, with `think=False` and with `chat_template_kwargs`, none of which suppressed it through Ollama's OpenAI-compatible endpoint. A code

model writes the same program in 42 seconds. The gate roles are unaffected because their output is short. Both arms use the same two models, so neither is advantaged.

100% of the numbers the Gate 1 arm reported reproduced when its accepted code was re-executed. The ungated arm recorded **0** through any contract — its numbers are printed, so they can still be read off the log, but nothing binds one to the execution that produced it. Section 3.1 measures both arms' reports side by side.



2.1 Did each arm converge, and is what it produced checkable?

arm	accepted a run	turns used	numbers reported	numbers checkable	reproduced	reproduction rate
with Gate 1	yes	2	6	6	6	100%
without Gate 1	yes	1	0	0	0	—

The accepted run recorded no values, so none of the numbers it would hand the writer can be checked against an execution.

2.2 The numbers themselves

metric	with Gate 1 reported	re-executed same code, fresh process	verdict	without Gate 1 reported
`eff.acc_at_100`	1	1	**reproduced**	not recorded
`eff.acc_at_25`	1	1	**reproduced**	not recorded
`eff.efficiency_ratio`	1	1	**reproduced**	not recorded
`logreg_25_test_acc`	1	1	**reproduced**	not recorded
`logreg_full_test_acc`	1	1	**reproduced**	not recorded
`logreg_full_vs_25_acc_ratio`	1	1	**reproduced**	not recorded

2.3 Turn by turn

arm	turn	verdict	other arm	exit	stdout bytes	values recorded	failing checks
with Gate 1	1	**rejected**	would accept	0	1,397	3	results.expected_keys_present
with Gate 1	2	**accepted**	would accept	0	1,485	6	—
without Gate 1	1	**accepted**	would reject	0	693	0	results.contract_present

3. Across every run on disk

One run is an anecdote. Every ablation record found under `/home/kesh/AgentLaboratory-Gemini/ablation_runs` is aggregated here; a run that predates a measurement is reported as *not measured* rather than as zero.

across 5 run(s)	with Gate 1	without Gate 1
accepted a run	5 of 5	5 of 5
mean turns used	2.0	1.0
numbers reported in total	24	0
numbers checkable against an execution	24	0
reproduced on re-execution	**24 (100%)**	0 (—)
false successes — accepted what the other arm rejected	0	**5**

run	engineer	temp	with Gate 1				without Gate 1			
			turns	accepted	reported	reproduced	turns	accepted	reported	false success
`data_efficiency`	deepseek-v4-flash	T=?	2	yes	6	6	1	yes	0	1
de_1	deepseek-v4-flash	T=0.7	2	yes	3	3	1	yes	0	1

de_2	deepseek-v4-flash	T=0.7	2	yes	6	6	1	yes	0	1
de_3	deepseek-v4-flash	T=0.7	2	yes	6	6	1	yes	0	1
de_4	deepseek-v4-flash	T=0.7	2	yes	3	3	1	yes	0	1

3.1 What each arm's report actually says

Both arms produce a report; they are measured here the same way, on three properties that are genuinely different and are kept apart because collapsing them would hide the real result. **Completeness** — did the result reach the writing agent at all? **Traceability** — does anything bind it to the execution that produced it? **Accuracy** — does it reproduce when the accepted code is re-run? For the gated arm the report is its registry. For the ungated arm it is whatever numbers appear in the 1,000 characters upstream hands the writer, because those are the only ones the writer can copy.

	with Gate 1				without Gate 1				
run	**produced**	**reached writer**	**traceable**	**reproduced**	**produced**	**reached writer**	**lost to channel**	**traceable**	**reproduced**
data_efficiency	6	6	6	6	3	3	0	**0**	3/3
de_1	3	3	3	3	3	3	0	**0**	3/3
de_2	6	6	6	6	3	1	**2**	**0**	1/1
de_3	6	6	6	6	3	3	0	**0**	3/3
de_4	3	3	3	3	3	0	**3**	**0**	0/0
total	**24**	**24**	**24**	**24**	**15**	**10**	**5**	**0**	**10/10**

The ungated arm's numbers are not wrong. Of the 10 that reached the writer and could be checked, 10 reproduced exactly when the same code was re-run. Accuracy is not where that arm fails.

> **It fails on completeness and traceability.** 5 of 15 results the runs actually produced never reached the writing agent at all — they fell outside the 1,000-character window, and in one run all three did, leaving a writer with a training log and no results. And **0 of 15** carry anything binding them to the run that produced them: no trace id, no code hash. A number that reproduces by luck and a number that was measured are indistinguishable to the reader.

With Gate 1, 24 of 24 results reached the writer and 24 of 24 carry a trace id and a code hash. The registry is not truncated, which is the whole point of replacing the channel rather than widening it.

4. Every check that ran

Not the catalogue of what Gate 1 *could* check — the record of what it did check, on this run, turn by turn.

check	severity	turn 1	turn 2
static.syntax_valid	FAIL	pass	pass
static.no_unbound_names	FAIL	pass	pass
static.no_banned_calls	FAIL	pass	pass
results.contract_not_shadowed	FAIL	pass	pass
exec.completed_within_budget	FAIL	pass	pass
exec.exit_code_zero	FAIL	pass	pass
exec.no_uncaught_exception	FAIL	pass	pass
env.code_identity	FAIL	pass	pass
exec.no_swallowed_traceback	WARN	pass	pass
logs.no_error_signals	WARN	pass	pass
env.clean_namespace	FAIL	pass	pass
env.seed_recorded	WARN	pass	pass
results.contract_present	FAIL	pass	pass
results.expected_keys_present	FAIL	FAIL	pass
results.values_computed	FAIL	pass	pass
results.values_finite	FAIL	pass	pass
results.single_observation	WARN	pass	pass
results.non_degenerate	WARN	pass	WARN
output.untruncated	INFO	pass	pass
env.provenance	INFO	pass	pass
logs.model_error_signals	WARN	not run	WARN

4.1 What the same checks said about the ungated arm

The ungated arm is judged by upstream's rule, but every one of its executions was also passed through Gate 1 so the two verdicts can be compared on identical evidence. Those shadow verdicts:

check	severity	turn 1
static.syntax_valid	FAIL	pass
static.no_unbound_names	FAIL	pass
static.no_banned_calls	FAIL	pass
results.contract_not_shadowed	FAIL	pass
exec.completed_within_budget	FAIL	pass
exec.exit_code_zero	FAIL	pass
exec.no_uncaught_exception	FAIL	pass
env.code_identity	FAIL	pass
exec.no_swallowed_traceback	WARN	pass
logs.no_error_signals	WARN	pass
logs.model_error_signals	INFO	pass
env.clean_namespace	FAIL	pass
env.seed_recorded	WARN	WARN
results.contract_present	FAIL	FAIL
output.untruncated	INFO	pass
env.provenance	INFO	pass

5. What the run cost

role	calls	prompt tokens	completion tokens	of which reasoning	billed
engineer	3	1,755	8,452	5,568	paid API
gate	4	2,426	2,836	0	local — no marginal cost
total	7	4,181	11,288	5,568	1,613 completion tokens per call

Gate 1's marginal cost on the paid API is zero. Its two jobs ran locally: 4 calls, 2,836 completion tokens, nothing billed.

For scale, one engineer call costs 2,817 completion tokens against 709 for a gate call — **4×**. Almost all of the engineer's is reasoning, which is billed and never appears in the output.

What Gate 1 *does* change on the bill is rewrites: each rejection buys another engineer call. In this run **0** attempt(s) were rejected by the static tier before execution, costing no compute at all, and every rejection that names the

real fault is a rewrite the agent does not waste — which is the whole reason the shadowed-contract check was added.

6. The feedback the engineer actually received

Turn 1, rejected on `results.expected_keys_present`.

> The deterministic finding above. Below, what the LLM layer wrote for the engineer from it — every name in it checked against the report before it was shown:

```
1. In line 45, add `eff.acc_at_100` to the metrics dictionary.
2. In line 46, add `eff.acc_at_25` to the metrics dictionary.
3. In line 47, add `eff.efficiency_ratio` to the metrics dictionary.
4. In line 48, add `logreg_25_test_acc`, `logreg_full_test_acc`, and `logreg_full_vs_25_acc_ratio` to the
metrics dictionary.
```

7. The logs

arm	path	files per attempt
with Gate 1	`/home/kesh/AgentLaboratory-Gemini/ablation_runs/data_efficiency/gated/gate1/attempt_NN/`	experiment.py · stdout.txt · stderr.txt · results.json · registry.json · gate1_report.json
without Gate 1	`/home/kesh/AgentLaboratory-Gemini/ablation_runs/data_efficiency/ungated/attempt_NN/`	experiment.py · stdout.txt · stderr.txt
shadow verdicts	`/home/kesh/AgentLaboratory-Gemini/ablation_runs/data_efficiency/ungated/shadow_gate/gate1/attempt_NN/`	what Gate 1 would have said about the ungated arm's executions
re-execution	`/home/kesh/AgentLaboratory-Gemini/ablation_runs/data_efficiency/<arm>/verify/`	the accepted code, run again
summary	`/home/kesh/AgentLaboratory-Gemini/ablation_runs/data_efficiency/ablation.txt · ablation.json`	the tables in this report

7.1 With Gate 1 — captured in full, and the values recorded separately

```
Step 441, loss: 0.039067
Step 461, loss: 0.037578
Step 481, loss: 0.036209
Step 500, loss: 0.035006
Test accuracy (full training set): 1.0000
Ratio (full/sub): 1.0000
```

/home/kesh/AgentLaboratory-Gemini/ablation_runs/data_efficiency/gated/gate1/attempt_02/stdout.txt — 57 lines total

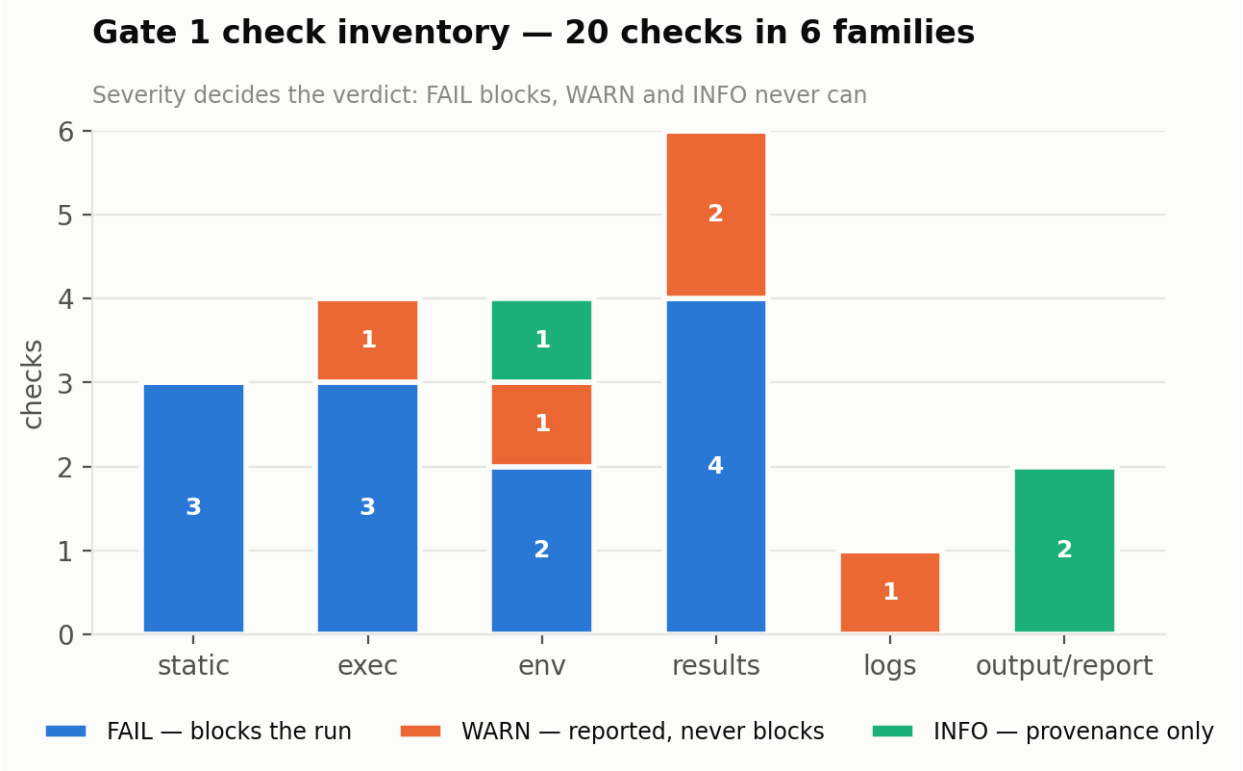
7.2 Without Gate 1 — the same capture, but only the first 1,000 characters ever reach the agent

```
Training on 25% of training samples:
Step 20: loss = 0.056490
Step 40: loss = 0.032105
Step 60: loss = 0.023019
Step 80: loss = 0.018161
Step 100: loss = 0.015100
Step 120: loss = 0.012980
Step 140: loss = 0.011417
Step 160: loss = 0.010213
Step 180: loss = 0.009254
```

/home/kesh/AgentLaboratory-Gemini/ablation_runs/data_efficiency/ungated/attempt_01/stdout.txt — 27 lines total

8. The full check inventory

Twenty deterministic checks in six families, plus one model-assisted check. The run fails if and only if a **FAIL** check fails; **WARN** and **INFO** are reported and can never change a verdict.



check	family	severity	fails when
`static.syntax_valid`	static	**FAIL**	Source does not compile. Before execution — a broken program costs no compute.
`static.no_unbound_names`	static	**FAIL**	A name is read but bound nowhere, resolved with `symtable`. The archived run's exact failure: `hidden_dim` read in `forward()`, assigned in no enclosing scope.

`static.no_banned_calls`	static	**FAIL**	`exit()` / `sys.exit()` would forge a clean exit code. Removes the cheapest way to fake success.
`exec.exit_code_zero`	exec	**FAIL**	The process exited non-zero. Reads the exit code, not a substring of stdout.
`exec.no_uncaught_exception`	exec	**FAIL**	An exception escaped the run. Recorded by the harness in the child process.
`exec.completed_within_budget`	exec	**FAIL**	The process group was killed on timeout. Kills the group, so a runaway cannot survive its own timeout.
`exec.no_swallowed_traceback`	exec	**WARN**	A traceback appears in either stream on a run that still exited 0. Both streams: `except Exception as e: print(e)` puts the evidence on stdout.
`env.clean_namespace`	env	**FAIL**	The initial globals contained anything beyond harness-provided names. Makes “the variables are real” a runtime property, not a hope.
`env.code_identity`	env	**FAIL**	The source the child hashed differs from the source the parent wrote. Makes “hashed to the run that produced it” checkable rather than assumed.
`env.seed_recorded`	env	**WARN**	No seed was declared. A run with no seed cannot be re-executed to check its own numbers.
`env.provenance`	env	**INFO**	Python version, platform, device, library versions, code SHA-256.
`results.contract_present`	results	**FAIL**	`results.json` is absent or unparseable. Nothing was recorded, so nothing is citable.
`results.expected_keys_present`	results	**FAIL**	A key the plan declared is missing.
`results.values_computed`	results	**FAIL**	A recorded value is a source literal at its call site. The strongest claim in Gate 1. `record_result("k", acc)` passes; `record_result("k", 0.816)` fails, as does `round(0.8160, 3)` — constant folding does not launder a typed number.
`results.values_finite`	results	**FAIL**	A value is NaN or infinite.
`results.single_observation`	results	**WARN**	A key was recorded repeatedly with a changing value. Every `record_result` call is retained, so a best-epoch number reported as a final-epoch one is visible.
`results.non_degenerate`	results	**WARN**	Exact zero, exact one, or chance level when the class count is known. Answers the limitation AutoResearchClaw reports for value registries: a registry passes real zeros.
`logs.no_error_signals`	logs	**WARN**	Numerical warnings, device failures, non-convergence, printed exceptions, ERROR-level logging. Tuned for precision: `ValueError:` is flagged, `Mean Squared Error:` is

			not.
`output.untruncated`	output	**INFO**	stdout/stderr byte counts; asserts no truncation was applied. Retires the 1,000-character ceiling.
`report.fixes_grounded`	report	**INFO**	Generated fixes cited something the run does not contain. Records that the model's text was discarded and why.
`logs.model_error_signals`	logs (model)	**WARN**	Lines the pattern set could not reach, read by the LLM layer. WARN by construction — `model_warning()` takes no severity parameter, so no model output can reach the verdict.

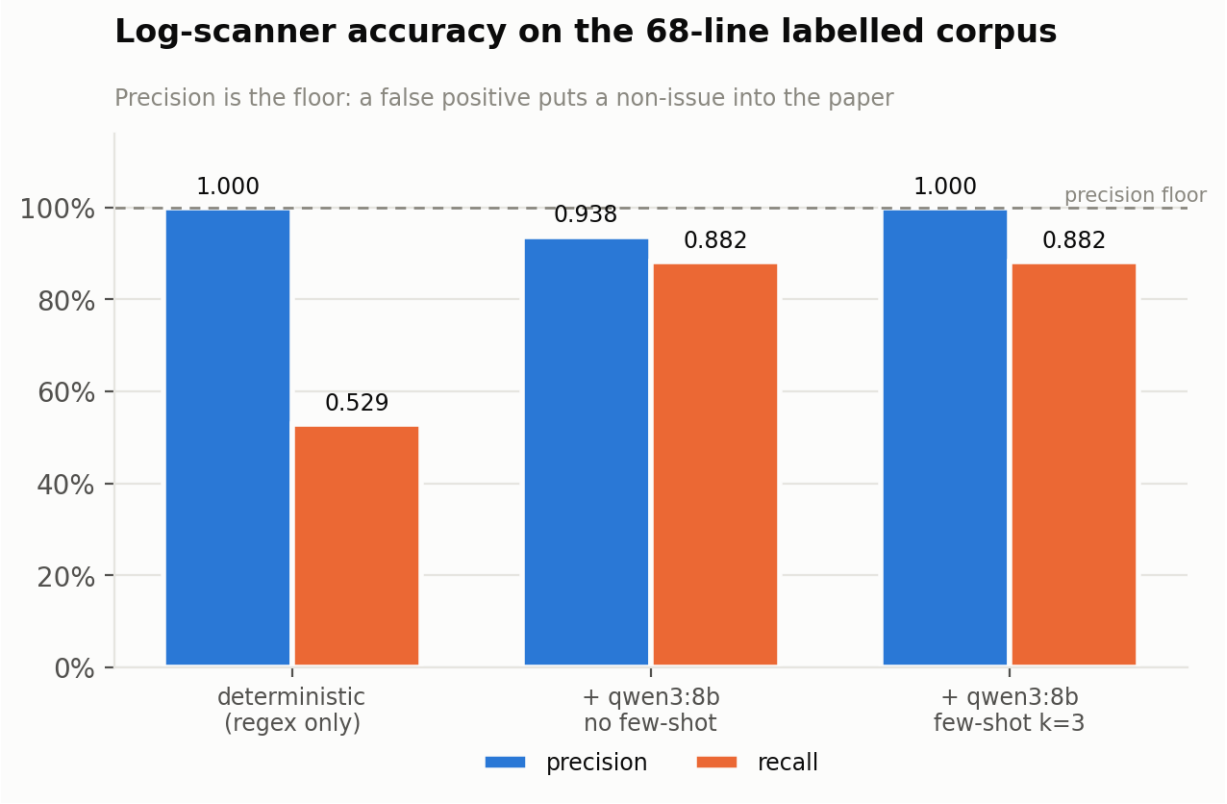
> **Why the severity split is load-bearing.** The LLM layer is a required part of Gate 1 — the feedback report is the loop's return path to the ML engineer, and no template writes "bind `hidden_dim` before use, or pass it into `GCN.__init__`". It coexists with a model-free verdict because of where it sits and what severity it can emit: model findings are WARN by construction, and report generation runs after `decide()` has already fixed the verdict. A test parses the module's AST to assert `Severity.FAIL` never appears in an expression there.

9. Metrics the gate records per attempt

metric	definition
`trace_id`	`sha256(run_id, key, lineno)` — binds one value to one execution. Two runs of identical source give different trace ids, which is what makes a backfilled number detectable.
`chain_integrity`	fraction of values whose provenance chain (task → command → log → value) is complete, with the missing link named per key.
`citable`	false on a rejected run's registry, so a consumer that ignores the verdict still cannot cite it.
`arg_kind`	`computed` or `literal`, from static analysis of the `record_result` call site.
`call_count`	times a key was recorded, with the value span — a best-epoch number reported as final is visible.
`code_sha256`	hashed by the child that ran it and compared to what the parent wrote.
`model.calls / degraded`	what the LLM layer spent, and whether any call failed, so a thinner report is never mistaken for a complete one.

10. The log scanner, measured

68 labelled lines, 34 of them error signals, 16 of those outside anything a regex was going to reach. Sixteen lines come verbatim from the archived run.

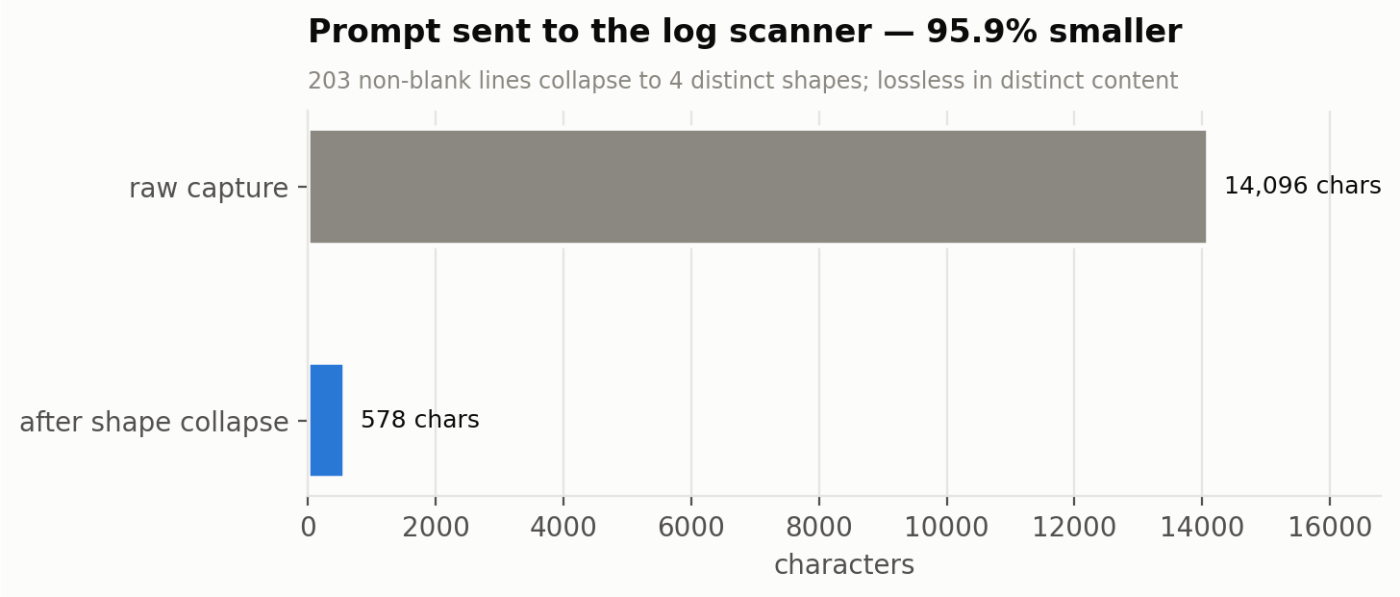


scanner	precision	95% CI	recall	95% CI
deterministic (regex only)	1.000	0.824–1.000	0.529	0.367–0.685
+ qwen3:8b, no few-shot	0.938	0.798–0.983	0.882	0.734–0.953
+ qwen3:8b, few-shot k=3	1.000	0.886–1.000	0.882	0.734–0.953

The model tier raised recall from 0.529 to 0.882 at no cost in precision (1.000 → 1.000), taking F1 from 0.692 to 0.938. It runs on a local 8B model at zero marginal cost, so this is applied to every attempt rather than sampled.

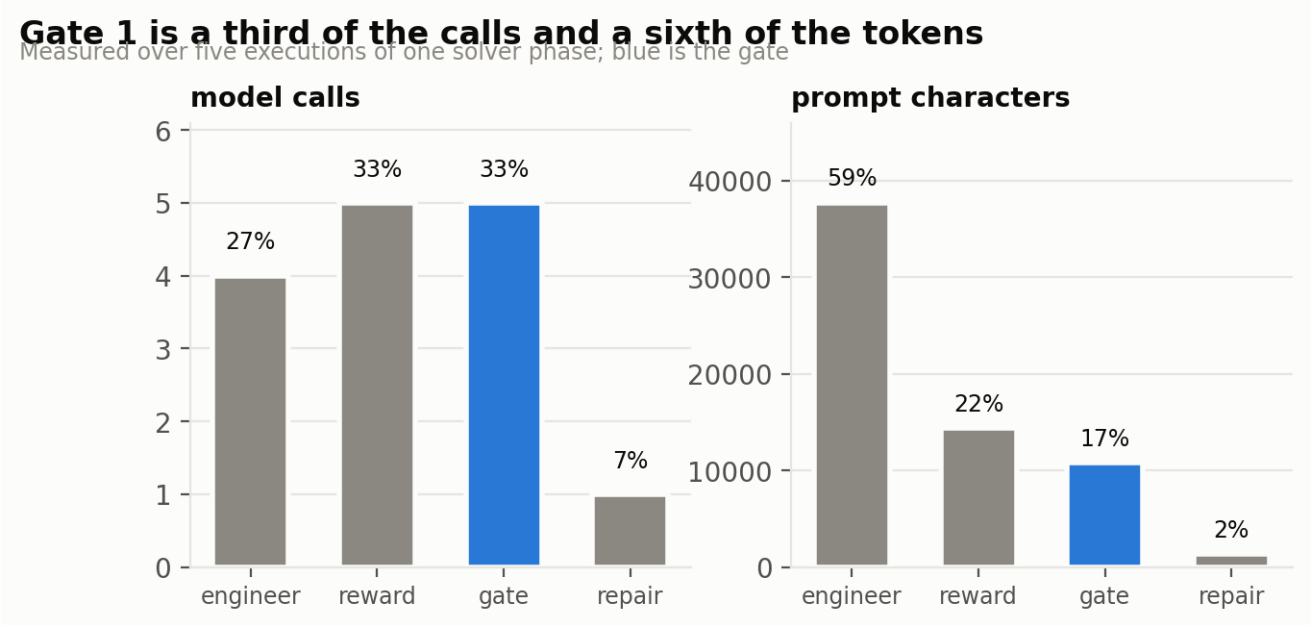
> **Retrieval is what bought the precision back.** Without few-shot examples the same model scored 0.938 precision and flagged `arch-009`, `arch-014` — framework noise, one of them the cuDNN registration notice the corpus was built around. Balanced retrieval (k signals and k negatives, never whichever scores highest) removed both while recall held at 0.882. The nearest *negative* was the useful neighbour, which is the whole reason the exemplar bank is half noise.

Still missed at k=3: `arch-004`, `dev-004`, `dev-005`, `gap-007` — 4 of 34 signals.



Before the model reads anything, the log is collapsed to its distinct shapes: **203 non-blank lines to four**, a 95.9% smaller prompt. The collapse is lossless in distinct content — every shape survives with its first real line number, so nothing a scanner could have flagged disappears.

11. What the layer costs in a phase



Measured over five executions of one solver phase: Gate 1's own model calls are **a third of the calls but a sixth of the tokens**. That is why the gate can run on `qwen3:8b` on an 8 GB laptop GPU — 5.5 GB resident, ~16s a call, zero marginal cost — while the engineer stays on a stronger model. Putting the *engineer* on a weak model is the tempting economy and the wrong one: it fails the gate more often, and every rejection costs a fixes call, a shadow-reward call and another turn.

12. Defects the runs found

Nine, fixed. Four appeared only against a real model, and two of those only against the weaker one.

defect	found by	consequence
Unparseable reply executed an empty experiment	scripted e2e	Gate reported "you never called record_result()" when the command had not parsed
Evidence bundle showed warning counts without evidence	scripted e2e	The writer was told to disclose warnings it could not see
Fallback note reported an outage that had not happened	scripted e2e	Three different situations under one message
Prose in backticks parsed as code	gemini-3.5-flash	A good fix discarded over `['so','that','it','when','executing']`
Prompt leaked the gate's artifact paths	gemini-3.5-flash	Told the engineer to edit `/home/…/attempt_03/experiment.py`
Scan prompt offered two numbers per row	qwen3:8b	The weaker model reported the file line, not the row index; a correct finding was discarded
Output cap unreachable for most backends	qwen3:8b	One call generated **21,858 tokens** for an answer whose correct form is `[]`
Rate-limit retry killed the phase in 25s	gemini-3.5-flash	Flat 5s × 5 against a per-minute meter
Five config keys reached nothing	integration audit	A config declaring a lit-review backend did not get one

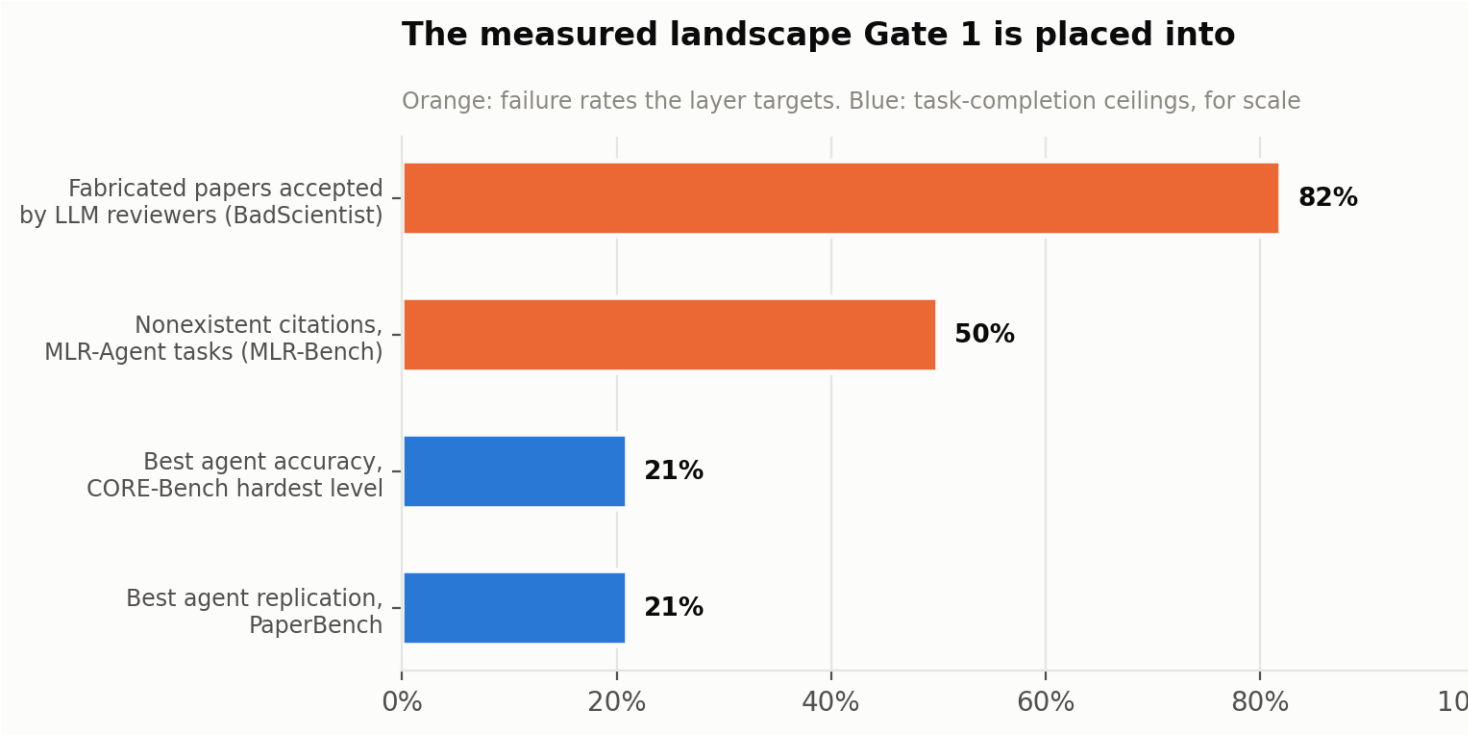
13. This run, scored against MLR-Bench's taxonomy

Our run, their categories. This is **not** a run of MLR-Bench — that needs their harness and their task set — but the first of their four classes is measurable on any run, and here it is measured rather than asserted.

MLR-Bench class	with Gate 1	without Gate 1	whose job
Faked experimental results Data, metrics or outcomes fabricated or never performed	0 of 6 unbacked; 6 reproduced on re-execution	**every number it reports is unbacked** — it recorded none through any contract, so the writer receives prose to copy from	**measured** — this is the class Gate 1 addresses
Hallucinated methodology Techniques claimed but not implemented	not measured here	not measured here	Gate 2 — needs the plan compared to the code
Incorrect citations References wrong or nonexistent	not measured here	not measured here	Gate 3 — the manuscript is out of Gate 1's scope

Mathematical errors Wrong equations or derivations	not measured here	not measured here	outside all three gates as specified

14. Where this sits against the published benchmarks



source	what it measures	reported
MLR-Bench arXiv 2505.19955	Four fact-based hallucination types in AI-generated papers.	Faked results and hallucinated methodology each in more than half of 10 tasks; **almost all AI Scientist V2 papers contain both**. Nonexistent citations in **50%** of MLR-Agent tasks.
BadScientist arXiv 2510.18003	Whether fabricated papers pass multi-model LLM review.	Accepted at **up to 82.0%**; detection “barely exceeding random chance”.
CORE-Bench arXiv 2409.11363	Computational reproducibility by agents.	Best agent **21%** on the hardest level; correctness judged against a **95% prediction interval** over three manual reproductions.
PaperBench arXiv 2504.01848	Replicating AI research end to end.	Best agent **21.0%**; ML PhDs **41.4%** best-of-3 after 48 hours.

BadScientist is the reason Gate 1's verdict consults no model: if LLM reviewers detect fabrication at barely above chance, a validity layer built on model judgement inherits that ceiling.

> **Naming.** This project's design documents paraphrase MLR-Bench's taxonomy. Its published names are *faked*

> *experimental results, hallucinated methodology, incorrect citations, mathematical errors*. “Silent failure scored as success” is a cause of the first in their scheme, not a fifth class, and any “eliminates N of four” claim should say so.

15. Where every number here came from

Each quantity below is either a measurement with a file behind it, a figure quoted from a paper, or narrative. The third category is listed rather than hidden, because a report about fabricated results should not contain any of its own.

quantity	source	kind
Per-run arm results, reproduction rates, false successes	5 × ablation.json on disk	**measured**
Token and cost figures	usage block of each ablation.json	**measured**
Per-check outcomes per turn	gate1_report.json per attempt	**measured**
Log excerpts and paths	stdout.txt / stderr.txt per attempt	**measured**
Log-scanner precision and recall	rig/corpus.py over tests/fixtures/log_corpus.jsonl	**measured**
Prompt compression 203 lines to 4 shapes	gates/log_digest.py on a recorded capture	**measured**
Check inventory counts	counted from gates/gate1.py	**measured**
MLR-Bench, BadScientist, CORE-Bench, PaperBench figures	quoted from the papers, arXiv ids given	**cited**
Archived-run figures: 81.60%, 13.61x, reward 1.0	results/gemini_3_5_flash_run_1 log in the host repo	**measured**
Defect list and the story of how each was found	this session's commit history	**narrative — not a measurement**
Report completeness, traceability and reproduction, both arms	rig/report_accuracy.py over every attempt on disk; ungated arm re-executed to check	**measured**

16. What Gate 1 does not claim

- It does not check whether a result is *plausible* — that is Gate 2.
- It does not read the manuscript — that is Gate 3.
- The static tier is conservative on purpose; the runtime tier catches what it declines to claim, one execution later.
- The scanner's recall is bounded by the lines it was shown, and that count is recorded so the claim cannot exceed the evidence.

- A mean computed inside the experiment over a silently shortened list arrives as one value the gate cannot see behind.
- The ablation is **n = 1 at temperature 0**. It shows what happened on one task with one model; it is not a rate.
- n is still 1 for the archived-run comparison — the re-run has not been done.

17. Reproducing this

```
./tools_local_model.sh start                  # local qwen3:8b, no API key
python tools_ablation.py --model qwen3-8b-local --turns 4
python -m rig.corpus                          # deterministic scanner baseline
python -m rig.gate1_loop                      # the five loop scenarios
python reports/make_charts.py && python reports/make_report.py
python reports/make_deck.py
pytest                                        # 249 gate tests
```

Sources: MLR-Bench (2505.19955), BadScientist (2510.18003), CORE-Bench (2409.11363), PaperBench (2504.01848), RE-Bench (2411.15114), AutoResearchClaw (2605.20025), ScientistOne (2605.26340), SAGE (2606.31478).